SA Node Action Potential *Prodigy Anesthesia*

An SA node action potential contains three distinct phases:

Phase 4: the spontaneous depolarization that triggers the action potential once the transmembrane potential reaches threshold (about -45 millivolts).

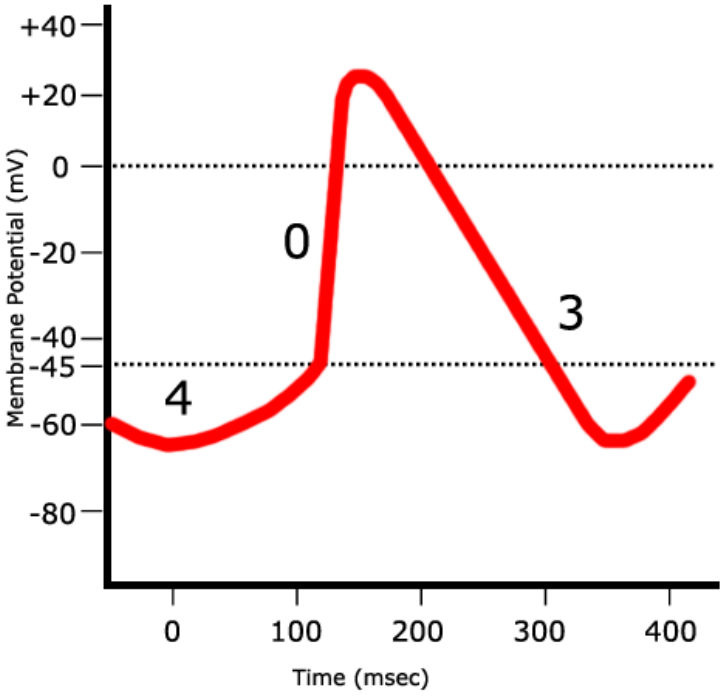
Phase 0: depolarization phase of the action potential

Phase 3: repolarization

At the end of repolarization (bottom of phase 3 and beginning of phase 4), ion channels open that allow the slow leakage of sodium into the cell. Once the membrane potential reaches about -45 mV, additional ion channels open that allow the flow of calcium into the cell. Additionally, this begins the decrease in potassium conductance out of the cell which contributes to the depolarizing of the membrane potential.

The depolarization phase (phase 0) is carried primarily by an increase in the conductance of calcium ions into the cell. Because the ion channels responsible for this are not very rapid, the rate of calcium influx, and therefore the slope of phase 0 is not as steep as that found in other cells such as Purkinje cells.

The repolarization phase (phase 3) occurs as outward-directed potassium channels open, allowing potassium to exit the cell quickly as the entry of calcium into the cell slows



Nagelhout JJ, Plaus K. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 478.

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